

Sentiment analysis project on movie and product reviews from people



1 Dataset collection and preprocessing for review sentiment



Gather diverse reviews with metadata for robust sentiment signals



Source quality checks and deduplication planning

Define inclusion rules, detect near-duplicates early, and track origin URLs to enable reliable reproducibility.

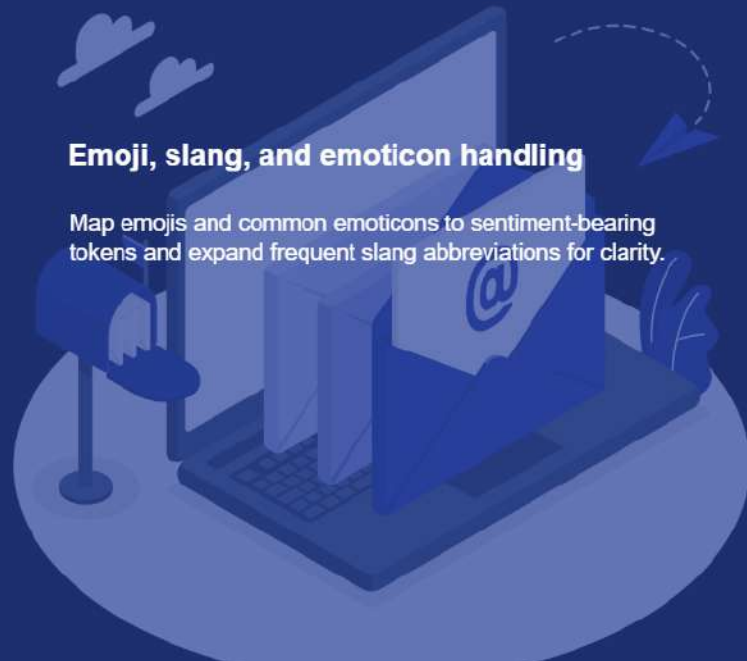
Clean text by removing noise,emojis,and duplicate content

Text normalization and noise reduction pipeline

Lowercase, remove HTML artifacts, standardize contractions, and normalize punctuation to reduce superficial variance.

Emoji, slang, and emoticon handling

Map emojis and common emoticons to sentiment-bearing tokens and expand frequent slang abbreviations for clarity.



Label examples using ratings and human guidelines for accuracy



Human annotation guidelines for edge cases

Create rules for sarcasm, mixed sentiment, and negation, then double-label ambiguous samples for agreement.



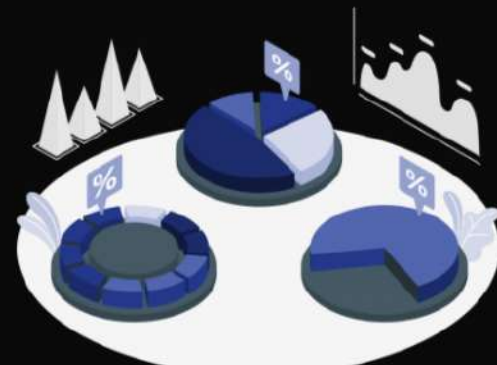
Rating-to-sentiment mapping with thresholds

Convert star ratings into sentiment classes using calibrated boundaries and validate mappings against observed review text.

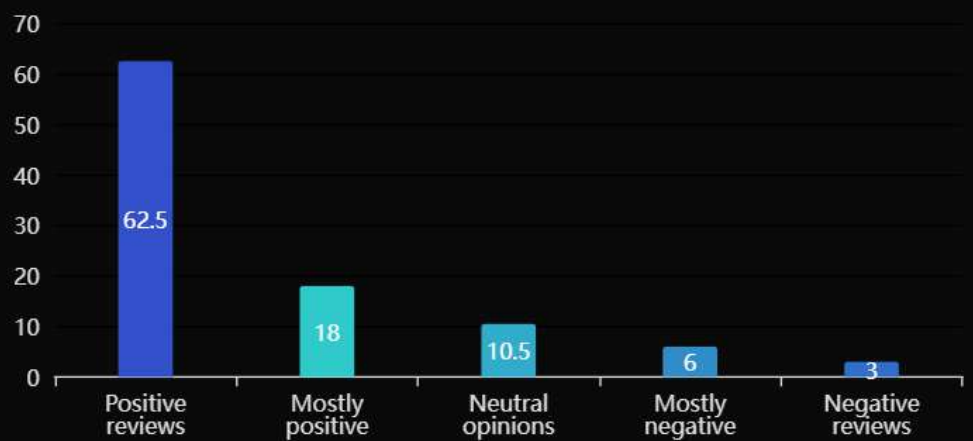


Continuous improvement through error analysis

Run model baselines, inspect misclassifications, and iteratively refine label guidelines for recurring failure patterns.



Sentiment Share For Movie Reviews



Movie or product review sentiment distribution(Percent)

2 Modeling, evaluation, and insights for review sentiment



Train classifiers using transformers and traditional baselines for comparison

Dataset creation from movie and product reviews

Collect labeled and unlabeled reviews, normalize text, and split into train validation test sets for robust modeling.

Evaluation-ready training pipeline and reproducibility

Use consistent seeds, versioned data, and checkpointing to ensure replicable results across baseline and transformer runs.



Text preprocessing and feature baselines setup

Apply tokenization, remove noise, and build TF-IDF, lexicon features, and logistic regression baselines for contrast.

Transformer fine-tuning for sentiment prediction

Fine-tune a pretrained transformer for positive, neutral, and negative sentiment using careful learning-rate scheduling.



Evaluate with precision recall and confusion matrix for reliability



Compute precision recall and F1 by class

Report per-class precision, recall, and F1 scores to quantify performance differences across sentiment categories.



Confusion matrix analysis for common mislabels

Inspect confusion matrices to identify frequent confusions, such as neutral predicted as positive in short reviews.



Threshold tuning and calibration for decision quality

Adjust decision thresholds and evaluate calibration to reduce overconfident predictions on ambiguous review text.



Error analysis for bias and domain drift

Review misclassified examples, check subgroup performance, and detect domain drift between movie and product domains.



Summarize themes and drivers behind positive and negative feedback



Feature attribution for sentiment-relevant terms

Apply attention visualization or explanation methods to link sentiment to meaningful phrases and product attributes.



Topic extraction for recurring satisfaction drivers

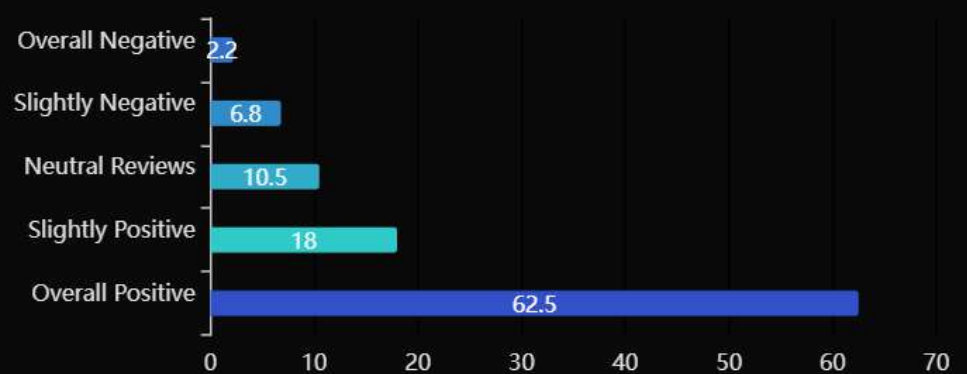
Use clustering or topic models to find recurring positive themes like acting quality, durability, or customer support.



Contrastive theme comparison across sentiment

Compare topics between positive and negative reviews to reveal what changes most under different sentiment labels.

Sentiment Overview for Movie Reviews



Movie review sentiment distribution(Percentage)

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Thank You